

21EIC553J - ROBOT PROGRAMMING

Submitted by

PALAK TIWARI - RA2512053010011

in partial fulfillment of the requirements for the degree of

MASTER OF TECHNOLOGY

in

AUTOMATION AND ROBOTICS



DEPARTMENT OF ELECTRONICS AND INSTRUMENTATION

COLLEGE OF ENGINEERING AND TECHNOLOGY

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

KATTANKULATHUR – 603203

MAY 2026

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

KATTANKULATHUR – 603203

BONAFIDE CERTIFICATE

This is to Certify that the laboratory work titled “ **ROBOT PROGRAMMING [21EIC553J]** ” is a Bonafide work carried out by **PALAK TIWARI - RA2512053010011**, who has successfully completed the laboratory requirements in partial fulfillment for the award of the degree of Master of Technology in Automation and Robotics during the academic year 2025-2026 [EVEN SEM].

**Dr. Xavier Reni
Prasad A
Associate Professor,
Department of Electronics and
Instrumentation,
School of Electrical and
Electronics, SRMIST.**

**Dr. Joselin Retna
Kumar G
Professor And Head,
Department of Electronics and
Instrumentation,
School of Electrical and
Electronics, SRMIST**

TABLE OF CONTENTS

Ex. No	Date	Title	Page No.	Signature
1.	28/01/2026	Alphanumeric and Symbolic Trajectory Synthesis	01	
2.	04/02/2026	Kinematic Alignment Logic and Multi-Point Spatial Mapping	05	
3.	18/02/2026	Dynamic Path Synthesis and Constant Velocity Control	09	
4.	25/02/2026	Discrete Matrix Palletizing	13	
5.	05/03/2026	Deterministic Calibration and Encoder Synchronization	17	
6.	12/03/2026	Autonomous Material Handling and Manipulation Logic	21	
7.	07/04/2026	Dual-Grid Synchronization and Sequential Routing Logic	25	
8.	15/04/2026	Process Orchestration for Multi-Stage Manipulation	29	